LAMMPS official dem04-hertz-oblique

Hertz wall impact: conditional saturated friction and spin | 29 October 2026

PASS	Four main runs evaluated using the unchanged official selected-time analytic criteria; source tag granular.

Official quantities, errors and acceptance limits

Quantity	Reference / limit	CPU dt	GPU dt	CPU dt/2	GPU dt/2
Rebound vz (m/s)	2.927848 relative limit 0.01	2.927842 err 1.943e-06 PASS	2.927842 err 1.943e-06 PASS	2.927849 err 4.609e-07 PASS	2.927849 err 4.609e-07 PASS
Rebound vx (m/s)	1.962679 relative limit 0.01	1.962634 err 2.275e-05 PASS	1.962634 err 2.275e-05 PASS	1.962634 err 2.278e-05 PASS	1.962634 err 2.278e-05 PASS
Rebound omega_y (rad/s)	544.2212 relative limit 0.01	541.768 err 0.004508 PASS	541.768 err 0.004508 PASS	541.768 err 0.004508 PASS	541.768 err 0.004508 PASS

Check time=3.317e-05 s; original step 3317. Half dt uses twice the check step at the same time. Each cell shows the native value and its own absolute/relative error; a reference limit is not a same-code epsilon.

Original bodies and source-to-native mapping

Body	Radius / density	Initial xyz (m)	Initial v (m/s)	Initial spin (rad/s)
sphere	R=0.0025 rho=2700	(0, 0, 0.00255)	(2.5069, 0, -2.9876)	(0, 0, 0)

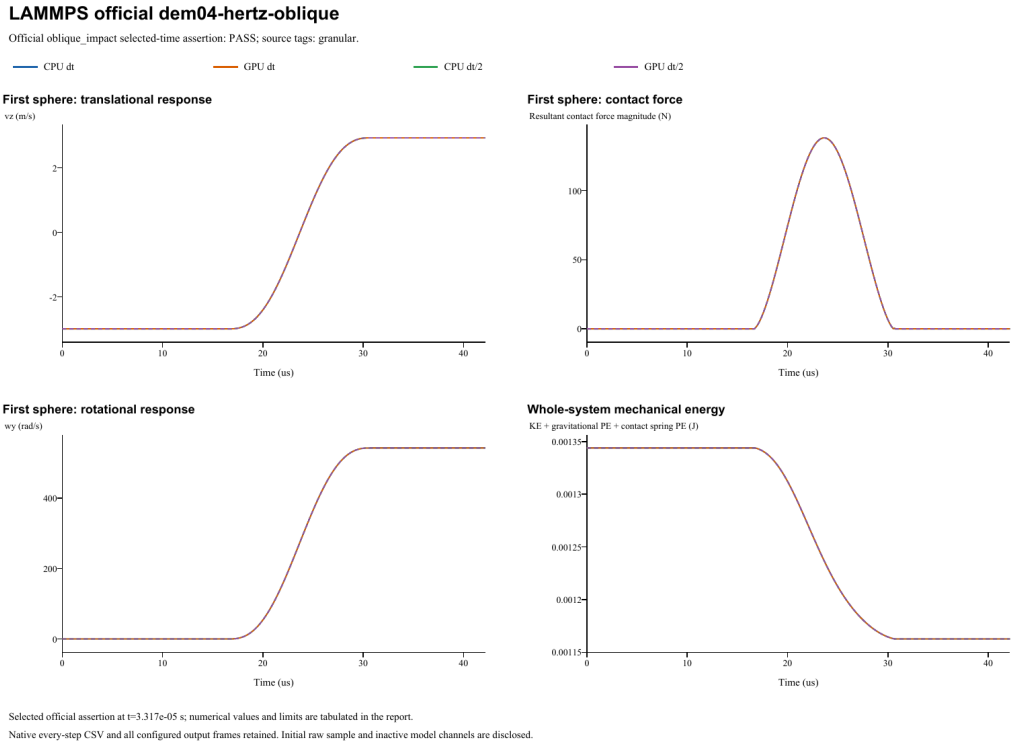
Hertz Fn_elastic=(4/3)*E_eff*sqrt(R_eff)*delta^1.5; native E_eff=3.9277275e+10 Pa. Stored linear kn is inactive. Automatic damping uses the harmonic material restitution and the active normal/tangential stiffness; stored manual dashpots are inactive. Tangential inputs: kt=0 N/m (explicit in linear mode), mu=0.092. Rolling/twist coefficients: kr=0, mu_r=0, ktwist=0, mu_twist=0. Source wall spin-slip and torque use R; native uses the shared contact point at R-delta. Maximum source/native contact-speed difference is 0.00512712 m/s; torque-arm difference is 1.40351e-05 m. Official selected-time acceptance does not imply pointwise kinematic identity. Source constant kt=1.5e10 N/m and ct=0 differ from native overlap-dependent Hertz/Mindlin stiffness and automatic damping. Their total Ft agrees only under the proved common saturation. Halving dt cannot remove the R versus R-delta torque convention.

Gravity=(0, 0, 0) m/s^2. Base/half dt=1e-08/5e-09 s; 4217/8434 steps. The four runs are accepted only for their independently proved common-Coulomb-cap path. Unsaturated tangential stiffness, damping and history operators are not validated. Direct Hertz conversion uses E_eff=0.75kn. Native nu=0.25 is an explicit unfitted encoding choice because the source direct-kn law supplies no Poisson ratio.

Pinned LAMMPS source

Official input | Analytic checker | Parameter mapping.
Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; input blob 5b07ff13041a03aff78974ac92de06db51eafcc0. Official tag: granular. LAMMPS was not rerun locally. Physical inputs were not fitted to native results.

LAMMPS native histories and refinement



CPU/GPU and half-step comparison

Quantity	CPU dt error	CPU dt/2 error	Error type / limit
Rebound vz (m/s)	1.9427e-06	4.6093e-07	relative: 0.01
Rebound vx (m/s)	2.2751e-05	2.2777e-05	relative: 0.01
Rebound omega_y (rad/s)	0.0045078	0.0045077	relative: 0.01

CPU/GPU whole-history maximum position / velocity / spin / force differences: 6.78e-21 m / 1.11e-15 m/s / 2.84e-13 rad/s / 6.42e-12 N.

Maximum individual impulse / position-drift / velocity-kick residuals: 3.05e-18 kg m/s / 2.17e-19 m / 4.38e-16 m/s. Gravity is added separately to saved native contact forces.

Source-force replay	Contact rows	Max Ft difference (N)	Max Fn difference (N)
CPU dt	1397	5.329e-15	5.684e-14
GPU dt	1397	5.329e-15	5.684e-14
CPU dt/2	2793	7.105e-15	5.684e-14
GPU dt/2	2793	5.329e-15	5.684e-14

Every contact has common source/native Coulomb saturation, including signed unloading. This proves total Ft only on the tested path. Spin error remains about 0.4508% at either timestep; half dt cannot remove the R versus R-delta torque convention. Unsaturated kt/Ct/history is outside scope.

Complete results and verified scope

All 25,306 native frames are retained; main frame counts=[4218, 4218, 8435, 8435]. Every numerical-step particle/contact CSV, four loadable main projects and raw initial samples remain preserved. Output cadence is recorded separately from integration dt. Energy/history diagnostics do not replace the official acceptance quantities.

Run manifest | Analysis | Official metrics | Independent audit | Full native ZIP | Frame hashes

Source checkpoints: Source checkpoint states are supplemental regression diagnostics. Their differences are reported directly; the YAML same-code epsilon is not imposed as an independent cross-software acceptance limit.

A stable helper executable is statically linked to the current native CPU/CUDA libraries. Historical GUI/library byte identity is not asserted. Scope is the active source physics and the selected official criteria; inactive options remain outside this test.